Mirror World — Design v4 (Simple Rules)

Oct 24, 2026 · @Aryendra Pratap Singh

v4 replaces the game rules and all levels. Every level was checked by a solver: each has one real puzzle, and mashing keys doesn't solve it.

How to use this doc

Teammates: attach this doc to your Claude chat along with the Claude Playbook and say: "Apply design v4. I'm Person A" (or B or C) "and here are my current files." Then attach or paste your files.

Instructions for Claude:

- v4 **replaces the game rules, the controls, the level format, all levels and the Rules interface** in design v2, the Claude Playbook and Design Update v3. Where they differ, **this doc wins**.
- **Unchanged:** the team roles and file owners, Phaser 3.90 with plain JavaScript, the folder layout, the art style and colours, the story tone (mysterious and bittersweet, never scary), and the 6:30 feature freeze.
- Person A already has finished, tested files for `src/rules.js`, `rules-test.html` and every level file. Never rewrite those unless a test fails; build everything else around them.
- First tell this person in 3–5 short lines what changes for them, then work through their steps in **Changes for A, B and C**.
- Remove code for everything listed under **Removed**. Don't keep it around "just in case".

Why v4

What was wrong: v3 had a dozen systems (lag, disobeys, Look, Will, Sam's route, Mercy, a crack counter), but the puzzles themselves were weak. You could mash keys and win, and the rules were hard to explain.

What v4 does: a few simple rules, and every level is one real puzzle built from them.

- **Reversal in every move:** the other body does the opposite of every key, left–right **and** up–down.

- **Roles reverse through the levels:** cracks swap Theo through the glass, a little more each act, until he ends up as the reflection.
- **Solver-checked:** each level was designed by a program that found its shortest solution and the odds of winning by mashing keys.

Level	Shortest solution	Win chance mashing 300 keys
1	9 moves	1.2%
2	14 moves	1.2%
3	15 moves	0%
4	14 moves	0%
5	17 moves	0%
6	17 moves	0%
7	17 moves	0%
8	6 / 12 / 18 moves (three endings)	Take Their Place 2.7%; the other two 0%

Mashing in Level 8 can only stumble into the villain ending, which fits the story.

Removed: lag, disobeys, the shimmer, Look, slip, the Undo lock, Mercy, Will, Follow / Resist / Break, Sam's route, Sam's Look, footprints, the Level 9 control card, the 0–5 crack counter, ONE MORE, strain, boxes, dark and swap tiles, shards, tile-triggered events, and Levels 9–12. The game is now **8 levels**.

The rules

The whole game is these seven rules.

1. **Your keys move Theo. The other body moves the opposite way:** press Right and it goes left; press Up and it goes down.
2. **Walls and closed doors block each body separately.** A blocked body stays put while the other still moves. This is how you line the two up.
3. **Plates open doors:** door A is open while anyone stands on plate a, or while someone is standing in the doorway (a door never closes on someone).
4. **Spikes:** if either body steps on spikes, the level fails.

5. **Cracks:** if either body steps on a crack, the two **swap places through the glass**. Each crack works once, then it's plain floor.
6. **Win:** Theo on his goal and the other body on theirs, at the same time.
7. **Final room:** stepping on the crack ends the game with Take Their Place; both standing on the mirror frame at once is Break It Together; both on their goals is Stay Behind.

And one rule of the world: **each room only ever holds one body.** A swap moves both, so Theo and the other are never in the same room.

Controls: arrows or WASD move · Z undo · R restart · M mute · Enter continues on story screens. Undo and restart work after a fail; after a win or an ending, keys are ignored until the next level loads.

What happens on each key press, in order:

1. Work out which doors are open right now.
2. Theo tries to move with the key; the other tries to move the opposite way. Each one that would hit a wall or a closed door stays put.
3. If either is on spikes, the level fails.
4. If either is on an unused crack, they swap places and the crack is used up.
5. Update the doors.
6. Check the win, or the endings in the final room.

Story and the role reversal

Same characters as before: Theo, Mara (the reflection, trapped long ago) and Sam (who moves in years later). The story is now carried by **where Theo stands**, not by extra systems.

Act	Levels	Where Theo is	What the player feels
1 — Obey	1–3	Always in the real room (left). The reflection does the opposite of him	"I'm in control; it just follows."
2 — Slip	4–6	Cracks pull him through the glass. Level 4: he must cross to reach his goal. Level 5: across and back. Level 6: across, and he doesn't come back	"The mirror is pulling me in."

Act	Levels	Where Theo is	What the player feels
— The swap	after 6	Swap scene: TOGETHER fades on the glass, Mara walks free, Theo is trapped	The turning point
3 — Behind the Glass	7–8	Starts in the mirror room (right). Sam, in the real room, now does the opposite of **him**	"Now I'm the reflection." From behind the glass, the real world is the one that copies you

The theme line for judges: first your reflection mirrors you, then the mirror swaps you, and in the end you are the reflection.

The three endings (Level 8)

Ending	How	Moves	What it means
Take Their Place	Step on the crack	6 (easiest)	Theo escapes and Sam is trapped. The cycle repeats
Stay Behind	Theo on his goal and Sam on hers	12	Theo stays a reflection so Sam can live a normal life
Break It Together (true ending)	Both stand on the mirror frame at once	18 (hardest)	The mirror shatters and everyone is free

The easy way out is the selfish one, and the true ending takes the most care. The crack is the quickest way out of the room, so the temptation is always in view.

The 8 levels

Person A has these as ready files (`src/levels/level01.js` to `level08.js`); `level09` – `level12` are now empty. Maps are here so everyone's Claude can see them.

Map symbols: `#` wall · `.` floor · `T` Theo · `O` the other (Mara in Acts 1–2, Sam in Act 3) · `H` Theo's goal · `D` the other's goal · `^` spikes · `a` plate · `A` its door · `%` crack · `F` mirror frame (Level 8). Left of `|` is the real room, right is the mirror room. Rooms are 7×6 with a wall border.

#	Room	Teaches	Story moment	Moves
1	Hallway	The other moves the opposite way; use a wall to stop one body. Theo is two steps from his goal, but that's not enough	The reflection blinks	9
2	Bathroom	Theo holds a plate so a door opens on the other side	I'M SORRY on the glass	14
3	Bedroom	Spikes: a safe step for Theo can be deadly for the reflection	The reflection waves	15
4	Study	The first crack. Theo's goal is behind the glass, so he has to cross	LET ME OUT	14
5	Kitchen	Two cracks: through the glass and back again	IT HAS TO BE SOMEONE	17
6	Attic	Plate, door and crack. Theo crosses and doesn't come back; the swap scene follows	The reflection tilts its head	17
7	Cellar, years later	Theo starts behind the glass; Sam does the opposite of him. Theo holds a plate to open Sam's door	IS SOMEONE THERE? on arrival; COME UPSTAIRS when solved	17
8	Attic, years later	The final choice	I CAN SEE YOU	6 / 12 / 18

Level 1 — Hallway

```
#######|#######
#..T.H#|#.#...#
##....#|#.D...#
#.....#|#.O...#
##....#|##....#
#######|#######
```

Solution: `L L D R D R R U R`

Level 2 — Bathroom

```
#######|#######
#..T..#|#...#O#
#.#...#|##D..A#
#a....#|#.....#
#H....#|#.....#
#######|#######
```

Solution: `D D L L U U D R R R R D L D`

Level 3 — Bedroom

```
#######|#######
#...H.#|#....##
#.....#|#^....#
#T....#|#O.^..#
#...#.#|#...D.#
#######|#######
```

Solution: `R R R R L D D R D L L U L U U`

Level 4 — Study

```
#######|#######
#D....#|#.....#
#..%..#|#...^O#
#..#..#|##H...#
#...T.#|#.....#
#######|#######
```

Solution: `U U L U U R U L L L L R D D` (the swap happens on move 3)

Level 5 — Kitchen

```
#######|#######
#.^#..#|#....D#
#.%...#|#...%.#
#.T^..#|#.....#
#..H..#|#.^O.##
#######|#######
```

Solution: `U U R R D D U L R R U L L D D D L`

Level 6 — Attic

```
#######|#######
#....D#|#^^..%#
#...T.#|#.#..##
#.....#|#...^O#
#..a#.#|#.H..A#
#######|#######
```

Solution: `D L D U R R D D D L L L D L D D L`

Level 7 — Cellar, years later

```
#######|#######
#D.^..#|#H..a.#
#A#...#|#.....#
#..O#.#|#...T##
#...^.#|#.....#
#######|#######
```

Solution: `D L U U R U L L R R D L L L D U U`

Level 8 — Attic, years later

```
#######|#######
##.#..#|#..#%^#
#D.FO.#|#.H^.F#
#.^...#|#.....#
#.....#|#.T..##
#######|#######
```

- Take Their Place: `R R R U U U`
- Stay Behind: `D R U R R D L L L L U R`
- Break It Together: `D R U R R D L L L U L L D R R R U R`

Contract v4

This replaces the **Level format** and **Rules interface** parts of the playbook's Shared contracts and v3's Contract changes. A's `src/rules.js` already follows it exactly.

Level format

```
LEVELS.push({
  number: 7, name: "Cellar, years later", act: 3,
  final: true,                  // Level 8 only
  map: [ "#######|#######", ... ],
  events: [
    { at: "start", do: "text", words: "IS SOMEONE THERE?", side: "left",
mirrored: true },
    { at: "win",   do: "text", words: "COME UPSTAIRS", side: "right",
mirrored: false }
  ],
  solution: "D L U U ...",      // Levels 1-7
  solutions: { take: "...", stay: "...", break: "..." }   // Level 8
});
```

- `events`: only `at: "start"` (as the level opens) or `at: "win"` (just before the win). `do` is `"text"` (with `words`, `side`, `mirrored`) or a moment: `"blink"`, `"wave"`, `"tilt"`.
- Symbols are listed in **The 8 levels**. `T` and `O` can be in either room.

Rules interface

```
Rules.load(level)          -> state
Rules.step(state, input)   -> { state, events }
```

Inputs: `"U"`, `"D"`, `"L"`, `"R"`, `"UNDO"`, `"RESTART"`. Nothing else. `step` never changes the state it's given.

State (what B reads):

Field	Meaning
`level`, `name`, `act`, `final`	From the level
`map`	The level's map, unchanged
`theo`	`{side, x, y}`: `side` is `"left"` or `"right"`; `x, y` are the column and row inside that half
`other`	`{side, x, y}` for Mara (Acts 1–2) or Sam (Act 3)
`usedCracks`	Cracks already used, as `"left:3:2"` strings. Draw these as floor
`openDoors`	Letters of the doors open now, like `["A"]`
`turn`	Moves so far
`status`	`"playing"`, `"won"`, `"failed"` or `"ended"` (Level 8)
`ending`	`null`, `"take"`, `"stay"` or `"break"`
`startEvents`	Events to play as the level appears

Events (in the order they happen):

Event	Fields	B should
`move`	`who` (`"theo"` or `"other"`), `to` `{side, x, y}`	Slide that sprite; `Effects.onMove(sprite)`
`blocked`	`who`	A small bump

Event	Fields	B should
`swap`	`at {side, x, y}` (the crack), `theo {...}`, `other {...}` (new places)	Move both sprites across the glass (a quick fade out and in), redraw that crack as floor, `Effects.onCrack(x, y)` at the crack's screen position
`door`	`letter`, `open`	Swap the door image
`fail`	`reason: "spikes"`	Pause 600 ms, then send `"RESTART"`
`text`	`words`, `side`, `mirrored`	Fade the words onto that side's glass, flipped if `mirrored`; `Effects.onMoment("text", null)`
`moment`	`kind`: `"blink"`, `"wave"`, `"tilt"`	`Effects.onMoment(kind, the other body's sprite)`
`win`	none	`Effects.onWin()`, then the next level. After Level 6, the swap scene first
`ending`	`which`	`Effects.onGlassBreak()` for `"take"` and `"break"`, then that ending's screen
`undo`, `restart`	none	Redraw everything straight from the new state, no slide

Art (C)

- **New image:** `crack.png` (32×32): a hairline crack on a floor tile, in silver `#9FD6E0`.
- **Goals:** B draws `goal.png` for both; tint Theo's goal (`H`) warm `#E9B872` and the other's (`D`) silver `#9FD6E0`, so players can tell which goal is whose.
- **Not needed any more:** `box.png`.
- The sprite for `other` is `reflection.png` in Acts 1–2 and `sam.png` in Act 3.

Changes for A, B and C

Person A — Logic (about 20 minutes, then testing and story)

A has a zip, `mirror-world-v4-A.zip`, holding the finished `src/rules.js`, `rules-test.html` and `src/levels/level01.js` – `level12.js`.

1. **Save first:** in GitHub Desktop, commit anything unsaved with the summary `before v4`, and push. Everything before v4 stays in GitHub's history.
2. **Tell B and C:** "Switching to design v4. Push what you have now, and don't touch any level file until I say it's in."
3. **Unzip into the `mirror-world` folder** and let it replace the files with the same names. Don't touch `src/levels.js`, `index.html` or anyone else's files.
4. **Test:** open `rules-test.html` with Live Server → **Check all levels**. All 10 lines must say PASS. Then play Level 1 by hand.
5. **Push** with the summary `Design v4: rules and 8 levels`, and tell B: "v4 rules and levels pushed. Pull now."
6. **Then, with B's OK, A takes over `src/story.js`** to save B time: rewrite the story cards for v4 (below). After that, play the live link start to finish after every push and send each owner a bug list.

Story cards to rewrite (`src/story.js`): remove `controlCard`. Keep `intro`, `act1`, `act2`, `act3`, `swap`, `together`, `endingTake`, `endingStay`, `endingBreak`, `credits`. Two to four short lines each, mysterious and bittersweet:

- `act2`: the glass has started to crack, and cracks pull you through.
- `swap` (after Level 6): Mara walks free; Theo is behind the glass.
- `act3`: years later, someone new lives here. From behind the glass, the real world is the one that copies you.
- `endingTake`, `endingStay`, `endingBreak`: one card each, matching the three endings.

Person B — Screens & story (about 60 minutes)

Pull A's v4 push first. Then change `src/game.js`:

1. **Draw each body in its own room**, using `state.theo.side` and `state.other.side`. Theo always uses `theo.png`. The other uses `reflection.png` in Acts 1–2 and `sam.png` in Act 3. Never assume Theo is on the left.
2. **New tiles:** `%` crack (`crack.png`, drawn as floor once it's in `state.usedCracks`) and `F` frame (`frame.png`). Goals: `H` tinted warm `#E9B872`, `D` tinted silver `#9FD6E0`.
3. **Keys:** arrows or WASD, Z, R, M, Enter. Remove Space and Shift.
4. **Events:** follow the events table in Contract v4, including the swap animation.
5. **Remove:** the Will meter, footprints, Sam's eye icon, the Level 9 control card, the Look hint, and every `Effects.setCracks` call. Bottom bar: "Arrows move · Z undo · R restart".

6. **Game flow:** title → `intro` → `act1` → Levels 1–3 → `act2` → Levels 4–6 → swap scene (`together` fades on the glass, then the `swap` card) → `act3` → Levels 7–8 → ending card → `credits` → title.
7. **Test:** play every level with the solutions in **The 8 levels**, plus all three Level 8 endings.

Done when: a full run from the title to each ending works with no red console errors. Then push and tell A and C.

Person C — Look & sound (about 30 minutes)

Change `src/effects.js` and `effects-test.html`:

1. **`crack.png`:** a new 32×32 floor tile with a silver hairline crack.
2. **`onCrack(x, y)`** now means "swap through the glass": a sparkle at that spot, a ripple along the mirror line and the crack sound. It's called once per swap.
3. **`onGlassBreak()`** plays for the Take Their Place and Break It Together endings.
4. **Remove** `setCracks` (no longer called) and the `shimmer`, `slip` and `strain` moments. Moments are now `blink`, `wave`, `tilt` and `text`.
5. **Colour fade by level:** Theo in full colour in Levels 1–3, fading through 4–6 while the reflection brightens, staying faded in 7–8.
6. **Music:** Theo's theme in Acts 1–2, Sam's theme in Act 3.
7. **Test page:** remove the Cracks 0–5, Shimmer, Slip and Strain buttons; add a **Swap** button that calls `onCrack` in the middle of the mirror line.

Done when: every button on `effects-test.html` works with no errors. Then push.

Order of work from now

Times count from the moment the team agrees to switch. The 6:30 feature freeze, the 7:30 final push and the 8:00 submission stay the same.

When	Who	Does	Message to send
+0 min	B, C	Push what you have now; don't edit level files	"Pushed, ready for v4"
+0–20	A	Steps 1–5: save, unzip, Check all levels, push	"v4 rules and levels pushed. Pull now."

When	Who	Does	Message to send
+0–30	C	All C changes; C never waits for anyone	"Effects updated for v4"
+20–80	B	Pull A's push, then all B changes	"Game updated for v4"
+20–80	A	`story.js` (once B agrees), then play-testing	"Story cards pushed"
After B's message	Everyone	Pull, play the live link start to finish, all three endings, and send bug lists to each owner	—

Who waits for whom: only B, for A's push at about +20 minutes. B can strip out the removed HUD pieces and keys while waiting.

If B is stuck after 60 minutes: A sits with B and helps debug `game.js`, but B stays the only one who edits it.

If time is very short: keep all 8 levels and the three endings. Cut, in this order: the swap animation (just redraw both bodies), the colour fade, Sam's sprite (reuse the reflection), the music themes.